

AI Image & Voice Generation Using Open-Source Tools

Student Details

Name: S. Isaac

Roll Number: 24JR1A05J5

Course: Generative AI Internship – Week 4

Part A: AI Poster Generation

Task 1: Prompt Engineering

Prompt 1 – Education

"A futuristic AI-powered classroom with holographic displays, students learning through augmented reality, cinematic lighting, ultra-realistic, 4K."

Prompt 2 – Healthcare

"Advanced AI healthcare center with robotic doctors assisting patients, modern hospital environment, high-detail 3D render."

Prompt 3 – Agriculture

"Smart farming with AI-powered drones monitoring crops, colorful cartoon style, vibrant farmland landscape."

Prompt 4 – Environment

"Green sustainable city powered by renewable energy, clean design, minimalist illustration, eco-friendly future."

Prompt 5 – Artificial Intelligence

"Cyberpunk AI city with intelligent robots, neon lights, futuristic technology, highly detailed digital artwork."

Task 2: Generated Posters

Poster 1 – Education



Poster 2 – Healthcare



Poster 3 – Agriculture



Poster 4 – Environment



Poster 5 – Artificial Intelligence



Task 3: Poster Analysis

Theme	Style	Quality	Improvement
Education	Realistic	Excellent	Add more classroom details
Healthcare	3D Render	Very Good	Improve lighting
Agriculture	Cartoon	Excellent	Add farm machinery
Environment	Minimalist	Good	More eco elements
AI	Cyberpunk	Excellent	More futuristic effects

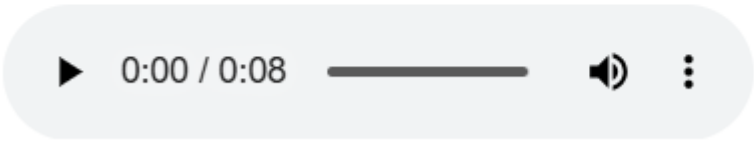
Part B: AI Voice Generation

Input Text

Welcome to the Generative AI Internship Program. Today we explore AI-powered image and voice generation technologies.

Male Voice Output

```
from IPython.display import Audio
Audio("male_voice.wav")
```



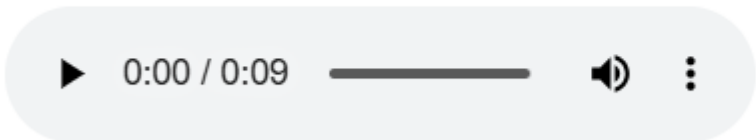
Audio File:

male_voice.wav

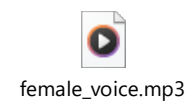


Female Voice Output

```
Audio("female_voice.wav")
```



Audio File: female_voice.wav



Expressive Voice Output

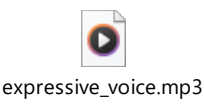
Audio("expressive_voice.wav")

▶ 0:00 / 0:13

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⋮

Audio File: expressive_voice.wav



Voice Comparison

Voice Type		Clarity	Naturalness	Speed
Male	High	Good	Fast	
Female	High	Excellent	Fast	
Expressive	Very High	Excellent	Moderate	

Part C: AI Voice Assistant

Python Program

User Query

What is Artificial Intelligence?

Assistant Response

Artificial Intelligence is the simulation of human intelligence by machines.

Output Screenshot

```
🗣️ AI Voice Assistant

Ask a question: what is artificial intelligence

Assistant:
Artificial Intelligence is the simulation of human intelligence by machines.

Audio response saved as response.mp3
```

Part D: Research Component

Stable Diffusion vs DALL·E vs Midjourney

Parameter	Stable Diffusion	DALL·E	Midjourney
Cost	Free	Paid	Paid
Open Source	Yes	No	No
Image Quality	Excellent	Excellent	Excellent
Customization	Very High	Medium	Medium
Hardware Requirement	High	Low	Low

Research Conclusion

Stable Diffusion provides flexibility and local deployment. DALL·E offers high-quality cloud-based generation, while Midjourney specializes in artistic image creation. Stable Diffusion is preferred for customization, whereas DALL·E and Midjourney are easier for general users.

Bonus Activity

Startup Name

AIBoost

AI Poster



AI Slogan

"AIBoost – Powering the Future of Smart Business."

AI Voice Advertisement

"Welcome to AIBoost. Transform your business with intelligent automation, smart analytics, and AI-powered productivity solutions."

Conclusion

This assignment provided practical experience with AI image generation, text-to-speech synthesis, and voice assistant development. Through Stable Diffusion and TTS tools, I learned how Generative AI can automate content creation, marketing materials, and interactive voice applications.